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Paper 32
Entered: August 20, 2025

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SAREPTA THERAPEUTICS, INC.,
Petitioner,

v.

THE TRUSTEES OF THE UNIVERSITY OF PENNSYLVANIA,
Patent Owner.

IPR2024-00580
Patent 11,680,274 B2

Before JEFFREY N. FREDMAN, ULRIKE W. JENKS, and
SUSAN L. C. MITCHELL, *Administrative Patent Judges*.

MITCHELL, *Administrative Patent Judge*.

JUDGMENT
Final Written Decision
Determining No Challenged Claim Unpatentable
35 U.S.C. § 318(a)

I. INTRODUCTION

Sarepta Therapeutics, Inc. ("Petitioner") filed a Petition (Paper 2, "Pet."), seeking *inter partes* review of claims 1, 3–6, and 8 of U.S. Patent No. 11,680,274 B2 (Ex. 1001, "the '274 patent"). The Trustees of the University of Pennsylvania ("Patent Owner") filed a Preliminary Response in which it raises challenges to the merits of the grounds in the Petition. Paper 8 ("Prelim. Resp."). On August 22, 2024, we granted institution of an *inter partes* review of claims 1, 3–6, and 8 of the '274 patent on all grounds set forth in the Petition. *See* Paper 9 ("Dec.") 2, 28.

Patent Owner filed a Response on December 5, 2024, *see* Paper 8 ("PO Resp."), Petitioner filed a Reply on March 20, 2025, *see* Paper 20 ("Reply"), and Patent Owner filed a Sur-Reply on April 24, 2025, *see* Paper 24 ("Sur-Reply"). An oral hearing was held on May 27, 2025, and a transcript of this hearing was entered into the record. Paper 31 ("Tr.").

During trial, Patent Owner stated that:

> To narrow the issues for the Board, Patent Owners have statutorily disclaimed claims 1 and 3–6 of the '274 Patent, leaving only claim 8 in this proceeding. Thus, Petitioner's Grounds 1–3 are no longer at issue apart from where Grounds 4–6 rely on them.

Sur-Reply 1, n.1. Thus, only Grounds 4–6 that challenge claim 8, are before us at this stage of the proceeding.

This is a Final Written Decision under 35 U.S.C. § 318(a) as to the patentability of the remaining challenged claim 8 on which we instituted trial. Based on the complete record before us, we determine as set forth below that the Petitioner has not shown by a preponderance of the evidence that claim 8 of the '274 patent is unpatentable.

II. BACKGROUND

A. Real Parties in Interest

Petitioner identifies Sarepta Therapeutics, Inc., Sarepta Therapeutics Three, LLC, and Catalent, Inc. as real parties in interest. *See* Pet. 5. Patent Owner identifies The Trustees of the University of Pennsylvania, GlaxoSmithKline LLC, REGENXBIO, and REGENXIO RS LLC, as the real parties in interest. *See* Paper 4, 1; Paper 7, 1; Paper 30, 1.

B. Related Matters

Petitioner and Patent Owner identify the following litigation as a related matter in which Petitioner is a defendant: *REGENXBIO Inc. f/k/a ReGenX v. Sarepta Therapeutics, Inc.*, C.A. No. 23-667-RGA (D. Del.) ("Penn-II"). Pet. 5; Paper 4, 2; Paper 7, 3.

C. The '274 Patent

The '274 patent issued on June 20, 2023, and is a divisional of an application filed February 18, 2015, now U.S. Patent No. 10,301,648, which itself is a divisional of an application filed on April 7, 2006, now U.S. Patent No. 8,999,678. Ex. 1001, codes (45), (62); 1:6–12. The '274 patent also claims priority to two provisional applications, the earliest of which was filed on April 7, 2005. *See id.* at code (60); 1:12–14.

The '274 patent relates to "a method of correcting singletons in a selected [Adeno-associated virus] AAV sequence in order to increas[e] the packaging yield, transduction efficiency, and/or gene transfer efficiency of the selected AAV," which involves "altering one or more singletons in the parental AAV capsid to conform the singleton to the amino acid in the corresponding position(s) of the aligned functional AAV capsid sequences." Ex. 1001, Abst.; 2:1–11; 2:63–3:3. The '274 patent notes that "AAV vectors have been described for use as delivery vehicles for both therapeutic and immunogenic molecules. To date, there have been several different well-characterized AAVs isolated from human or non-human primates (NHP)." *Id.* at 1:47–50. It is also noted that the literature reported that "different AAVs exhibit different transfection efficiencies, and also exhibit tropism for different cells or tissues." *Id.* at 1:57–59.

The method described in the '274 patent improves the function of an AAV vector by improving the packaging yield, transduction efficiency, and/or gene transfer efficiency of an AAV vector having a capsid of an AAV which contains one or more singletons. Ex. 1001, 2:1–4, 2:63–3:1, 3:32–35. The singleton is described as "a variable amino acid in a given position in a selected (*i.e.*, parental) AAV capsid sequence." *Id.* at 3:10–12.

> The "singleton" or variable amino acid may be found by aligning the sequence of the parental AAV capsid with a library of functional AAV capsid sequences. The sequences are then analyzed to determine the presence of any variable amino acid sequences in the parental AAV capsid where the sequences of the AAV in the library of functional AAVs have complete conservation. The parental AAV sequence is then altered to change the singleton to the conserved amino acid identified in that position in the functional AAV capsid sequences.

Ex. 1001, 3:13–21, 6:44–47; *see also id.* at 6:14–19 ("A singleton is identified where, for a selected amino acid position when the AAV sequences are aligned, all of the AAVs in the library have the same amino acid residue (i.e., are completely conserved), but the parental AAV has a different amino acid residue."). The replacement of the singleton with the conserved amino acid can be done by conventional site-directed mutagensis techniques. *Id.* at 6:44–50.

Although a parental AAV sequence may have more than six singletons, the ’274 patent states that “according to the present invention, a parental AAV sequence may have 1 to 6, 1 to 5, 1 to 4, 1 to 3, or 2 singletons.” Ex. 1001, 3:21–23, 6:62–63. The ’274 patent notes that improvement in function may be observed by correction of only one singleton. *Id.* at 6:63–64. The ’274 patent states:

> In the embodiment where a parental AAV carries multiple singletons, each singleton may be altered at a time, followed by assessment of the modified AAV for enhancement of the desired function. Alternatively, multiple singletons may be altered prior to assessment for enhancement of the desired function.
>
> Even where a parental AAV contains multiple singletons and functional improvement is observed altered of a first singleton, it may be desirable to optimize function by altering the remaining singleton(s).
>
> . . . .
>
> These altered AAVs have novel capsids produced according to the method of the invention and are assessed for function. Suitable methods for assessing AAV function have been described herein and include, e.g., the ability to produce DNAse protected particles, in vitro cell transduction efficiency, and/or in vivo gene transfer. Suitably, the altered AAVs of the invention have a sufficient number of singletons altered to increase function in one or all of these characteristics, as compared to the function of the parent AAV.

Ex. 1001, 6:65–7:7, 7:16–24.

The library of functional AAV capsid sequences used for comparison to a parental AAV to identify singletons “is characterized by a desired level of packaging ability, a desired level of in vitro transduction efficiency,

[and]/or a desired level of in vivo gene transfer efficiency (i.e., the ability to deliver to a target selected target tissue or cell in a subject).” Ex. 1001, 3:48–55. AAVs identified as suitable for use in the functional libraries include AAV1, AAV2, AAV6, AAV7, AAV8, AAV9, and other sequences described in two listed publications. *Id.* at 5:18–23.

The ’274 patent also describes how the novel AAV capsid sequences generated by mutation of the parental AAV at one or more of the singletons may be used to deliver a transgene, which encodes a polypeptide, protein, or other product of interest, to a host. *See* Ex. 1001, 12:7–9, 19:35–21:26. The ’274 patent states:

> In another aspect, the present invention provides a method for delivery of a transgene to a host which involves transfecting or infecting a selected host cell with a recombinant viral vector generated with the singleton-corrected AAV (or functional fragments thereof) of the invention. Methods for delivery are well known to those of skill in the art and are not a limitation to the present invention.

Ex. 1001, 19:35–41.

D. Challenged Claims

The Petition challenges claims 1, 3–6, and 8, *see* Pet. 1, 7–8, but, as previously stated, Patent Owner disclaimed claims 1 and 3–6, leaving claim 8 as the only remaining challenged claim before us. Because claim 8 depends ultimately from claim 1, and we will be discussing limitations from claim 1 that are also required for claim 8, we reproduce both claims 1 and 8 below.

> 1. A recombinant adeno-associated virus (AAV) comprising an AAV capsid and a minigene having AAV inverted terminal repeats and a heterologous gene operably linked to regulatory sequences which direct expression of the heterologous gene in a

> host cell, wherein the AAV capsid comprises AAV vp1 proteins, AAV vp2 proteins, and AAV vp3 proteins, wherein the AAV vp1 proteins have i) the sequence of amino acids 1 to 738 of SEQ ID NO: 4 (AAVrh46), or ii) an amino acid sequence at least 95% identical to the full length of amino acids 1 to 738 of SEQ ID NO:4, wherein the amino acid residue corresponding to position 665 in SEQ ID NO: 4 is N when aligned along the full length of amino acids 1 to 738 of SEQ ID NO: 4.

Ex. 1001, 193:2–14.

> 8. The recombinant AAV according to claim 6, wherein the dystrophin gene product is a mini-dystrophin or micro-dystrophin.

Ex. 1001, 194:24–26.

E. Asserted Grounds of Unpatentability

The remaining grounds alleging unpatentability of claim 8 that are asserted by Petitioner are listed below.

Claim(s) Challenged	**35 U.S.C. §[1]**	**Reference(s)/Basis**
8	103(a)	’772 Publication[2], Fabb[3]

[1] The Leahy-Smith America Invents Act, Pub. L. No. 112-29, 125 Stat. 284 (2011) (“AIA”), included revisions to 35 U.S.C. § 103 that became effective on March 16, 2013, after the filing of the applications to which the ’274 patent claims priority. Therefore, we apply the pre-AIA version of Section 103.

[2] Guangping Gao et al., US 2003/0138772 A1, published July 24, 2003 (Ex. 1007, “’772 Publication”).

[3] Stewart A. Fabb et al., *Adeno-associated virus vector gene transfer and sarcolemmal expression of a 144 kDa micro-dystrophin effectively restores the dystrophin-associated protein complex and inhibits myofiber degeneration in nude/mdx mice*, 11 Human Molecular Genetics 733–741 (2002) (Ex. 1010, “Fabb”).

Claim(s) Challenged	35 U.S.C. §[1]	Reference(s)/Basis
8	103(a)	’772 Publication, Xie,[4] Fabb
8	103(a)	’772 Publication, Snowdy,[5] Fabb

Pet. 7–8.

Petitioner further relies on the declarations of David V. Schaffer, Ph.D. (Exs. 1005, 1072). Patent Owner relies on the declaration of Richard Jude Samulski, Ph.D. Ex. 2027.

III. ANALYSIS OF THE ASSERTED GROUNDS

A. Legal Standards

"In an [*inter partes* review], the petitioner has the burden from the onset to show with particularity why the patent it challenges is unpatentable." *Harmonic Inc. v. Avid Tech., Inc.*, 815 F.3d 1356, 1363 (Fed. Cir. 2016) (citing 35 U.S.C. § 312(a)(3) (requiring *inter partes* review petitions to identify "with particularity . . . the evidence that supports the grounds for the challenge to each claim")). This burden of persuasion never shifts to the patent owner. *See Dynamic Drinkware, LLC v. Nat'l Graphics, Inc.*, 800 F.3d 1375, 1378 (Fed. Cir. 2015) (discussing the burden of proof in *inter partes* review).

A claim is unpatentable under 35 U.S.C. § 103(a) if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious before the

[4] Qing Xie et al., *The atomic structure of adeno-associated virus (AAV-2), a vector for human gene therapy*, 99 Proc. Nat'l Acad. Sci., 10405–10410 (2002) (Ex. 1008, "Xie").
[5] Stephen Snowdy, *Nuclear targeting by adeno-associated virus capsid proteins and virions* (2003) (Ph.D. dissertation, University of North Carolina, Chapel Hill) (Ex. 1009, "Snowdy").

effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. *See KSR Int'l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of ordinary skill in the art; and (4) objective evidence of nonobviousness, if any. *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

In analyzing the obviousness of a combination of prior art elements, it can be important to identify a reason that would have prompted one of skill in the art "to combine . . . known elements in the fashion claimed by the patent at issue." *KSR*, 550 U.S. at 418. A precise teaching directed to the specific subject matter of a challenged claim is not necessary to establish obviousness. *Id.* Rather, "any need or problem known in the field of endeavor at the time of invention and addressed by the patent can provide a reason for combining the elements in the manner claimed." *Id.* at 420.

Accordingly, a party that petitions the Board for a determination of unpatentability based on obviousness must show that "a skilled artisan would have been motivated to combine the teachings of the prior art references to achieve the claimed invention, and that the skilled artisan would have had a reasonable expectation of success in doing so." *In re Magnum Oil Tools Int'l, Ltd.*, 829 F.3d 1364, 1381 (Fed. Cir. 2016) (internal quotation marks omitted). "Both the suggestion and the expectation of success must be founded in the prior art, not in the applicant's disclosure." *In re Dow Chemical Co.*, 837 F.2d 469, 473 (Fed. Cir. 1988).

An obviousness analysis "need not seek out precise teachings directed to the specific subject matter of the challenged claim, for a court can take account of the inferences and creative steps that a person of ordinary skill in the art would employ." *KSR*, 550 U.S. at 418; *see In re Translogic Tech, Inc.*, 504 F.3d 1249, 1259 (Fed. Cir. 2007). In *KSR*, the Supreme Court also stated that an invention may be found obvious if trying a course of conduct would have been obvious to a person of ordinary skill in the art:

> When there is a design need or market pressure to solve a problem and there are a finite number of identified, predictable solutions, a person of ordinary skill has good reason to pursue the known options within his or her technical grasp. If this leads to the anticipated success, it is likely the product not of innovation but of ordinary skill and common sense. In that instance the fact that a combination was obvious to try might show that it was obvious under § 103.

KSR, 550 U.S. at 421. "*KSR* affirmed the logical inverse of this statement by stating that § 103 bars patentability unless 'the improvement is more than the predictable use of prior art elements according to their established functions.'" *In re Kubin*, 561 F.3d 1351, 1359–60 (Fed. Cir. 2009) (citing *KSR*, 550 U.S. at 417).

We analyze the asserted grounds of unpatentability in accordance with the above-stated principles. In making such an analysis, we find that Petitioner has not shown by a preponderance of the evidence that claim 8 of the '274 patent would have been unpatentable.

B. Level of Ordinary Skill in the Art

In determining the level of skill in the art, we consider the type of problems encountered in the art, the prior art solutions to those problems, the rapidity with which innovations are made, the sophistication of the technology, and the educational level of active workers in the field. *See Custom Accessories, Inc. v. Jeffrey-Allan Industries, Inc.*, 807 F.2d 955, 962 (Fed. Cir. 1986); *see also Orthopedic Equip. Co. v. United States*, 702 F.2d 1005, 1011 (Fed. Cir. 1983).

In addressing the level of ordinary skill in the art, Petitioner contends that a person of ordinary skill in the art ("POSA" or "POSITA") would have had

> at least a Ph.D. in biochemistry, molecular biology, or a related field and between one and four years of post-doctoral experience in the field of gene therapy. Alternatively, a POSA would have had at least a Master's or Bachelor's Degree in biochemistry, molecular biology, or a related field, with a corresponding number of additional years of experience in the field of gene therapy.

Pet. 12 (citing Ex. 1005 ¶¶ 111–115). Patent Owner states that it does not challenge Petitioner's definition. PO Resp. 10.

On this complete record, we accept Petitioner's proposed definition, as it appears consistent with the level of skill in the art reflected in the prior art of record and the disclosure of the '274 Patent. *See Okajima v. Bourdeau*, 261 F.3d 1350, 1355 (Fed. Cir. 2001) ("the prior art itself [may] reflect[] an appropriate level" as evidence of the ordinary level of skill in the art (quoting *Litton Indus. Prods., Inc. v. Solid State Sys. Corp.*, 755 F.2d 158, 163 (Fed. Cir. 1985)).

C. Claim Construction

We interpret a claim "using the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. 282(b)." 37 C.F.R. § 42.100(b). Under this standard, we construe the claim "in accordance with the ordinary and customary meaning of such claim as understood by one of ordinary skill in the art and the prosecution history pertaining to the patent." *Id.* Moreover, "the specification 'is always highly relevant to the claim construction analysis. Usually it is dispositive; it is the single best guide to the meaning of a disputed term.'" *In re Abbott Diabetes Care Inc.*, 696 F.3d 1142, 1149 (Fed. Cir. 2012) (quoting *Phillips v. AWH Corp.*, 415 F.3d 1303, 1315 (Fed. Cir. 2005) (en banc)).

Petitioner asserts that the '274 patent "does not specify a particular program or set of program parameters that should be used to align sequences and calculate percent identity for purposes of determining whether a particular test sequence falls within the scope of the claims." Pet. 21. Petitioner asserts that it analyzed the percent identity claim elements, as Patent Owner did in the parallel District Court litigation, Penn-II, "using the Clustal program at default settings to generate amino acid sequence alignments and calculate percent identity values." *Id.* at 21–22. Petitioner, however, does not ask for any particular construction of any claim term. *See id.*

Patent Owner states that it "does not challenge Petitioner's articulation of the lack of a need for claim construction." PO Resp. 10.

Having considered the record, we determine that no express claim construction of any claim term is necessary to reach our decision. *See Realtime Data, LLC v. Iancu*, 912 F.3d 1368, 1375 (Fed. Cir. 2019) ("The Board is required to construe 'only those terms that . . . are in controversy, and only to the extent necessary to resolve the controversy.'") (quoting *Vivid Techs., Inc. v. Am. Sci. & Eng'g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999)); *Nidec Motor Corp. v. Zhongshan Broad Ocean Motor Co. Ltd. v. Matal*, 868

F.3d 1013, 1017 (Fed. Cir. 2017) ("[W]e need only construe terms 'that are in controversy, and only to the extent necessary to resolve the controversy.'" (quoting *Vivid Techs.*, 200 F.3d at 803)).

D. Asserted Obviousness Grounds for Claim 8

Petitioner asserts three grounds that challenge claim 8 based on the '772 Publication and Fabb, as well as in combination with Xie or Snowdy. *See* Pet. 7–8. We determine that Petitioner has failed to prove by a preponderance of the evidence that claim 8 would have been unpatentable based on any of these three grounds because Petitioner has not shown why one of ordinary skill in the art would combine the teachings of the asserted references to arrive at the claimed invention with a reasonable expectation of success.

We begin with our review of the pertinent teachings of the asserted art.

i. '772 Publication (Ex. 1007)

The '772 Publication describes

> a method which takes advantage of the ability of adeno-associated virus (AAV) to penetrate the nucleus, and, in the absence of a helper virus co-infection, to integrate into cellular DNA and establish a latent infection. This method utilizes a polymerase chain reaction (PCR)-based strategy for detection, identification and/or isolation of sequences of AAVs from DNAs from tissues of human and non-human primate origin as well as from other sources.

Ex. 1007 ¶ 15. The '772 Publication also provides nucleic acid sequences identified by this method, such as serotype 7 or AAV7, AAV10, AAV11, and AAV12. *Id.* ¶ 16.

The ’772 Publication further discloses:

> FIG. 1 provides the non-human primate (NHP) AAV nucleic acid sequences of the invention in an alignment with the previously published AAV serotypes, AAV1 [SEQ ID NO:6], AAV2 [SEQ ID NO:7], and AAV3 [SEQ ID NO:8]. These novel NHP sequences include those provided in the following Table 1, which are identified by clone number:

TABLE 1

AAV Cap Sequence	Clone Number	Source Species	Tissue	SEQ ID NO (DNA)
Rh.1	Clone 9 (AAV9)	Rhesus	Heart	5
Rh.2	Clone 43.1	Rhesus	MLN	39
Rh.3	Clone 43.5	Rhesus	MLN	40
Rh.4	Clone 43.12	Rhesus	MLN	41
Rh.5	Clone 43.20	Rhesus	MLN	42
Rh.6	Clone 43.21	Rhesus	MLN	43
Rh.7	Clone 43.23	Rhesus	MLN	44
Rh.8	Clone 43.25	Rhesus	MLN	45
Rh.9	Clone 44.1	Rhesus	Liver	46
Rh.10	Clone 44.2	Rhesus	Liver	59
Rh.11	Clone 44.5	Rhesus	Liver	47
Rh.12	Clone 42.1B	Rhesus	MLN	30
Rh.13	42.2	Rhesus	MLN	9
Rh.14	Clone 42.3A	Rhesus	MLN	32
Rh.15	Clone 42.3B	Rhesus	MLN	36
Rh.16	Clone 42.4	Rhesus	MLN	33
Rh.17	Clone 42.5A	Rhesus	MLN	34
Rh.18	Clone 42.5B	Rhesus	MLN	29

Ex. 1007 ¶ 66 (reproducing a partial table showing clones Rh.1 to Rh.18). Of particular interest for this proceeding is Rh.10 or Clone 44.2. *See* Pet. 13–14.

In describing recombinant viruses and uses of them, the ’772 Publication states:

> Using the techniques described herein, one of skill in the art may generate a rAAV having a capsid of a novel serotype of the invention, or a novel capsid containing one or more novel fragments of an AAV serotype identified by the method of the invention. In one embodiment, a full-length capsid from a

> single serotype, e.g., AAV7 [SEQ ID NO: 2] can be utilized. In another embodiment, a full-length capsid may be generated which contains one or more fragments of a novel serotype of the invention fused in frame with sequences from another selected AAV serotype. For example, a rAAV may contain one or more of the novel hypervariable region sequences of an AAV serotype of the invention. Alternatively, the unique AAV serotypes of the invention may be used in constructs containing other viral or non-viral sequences.
>
> It will be readily apparent to one of skill in the art . . . that certain serotypes of the invention will be particularly well suited for certain uses. For example, vectors based on AAV7 capsids of the invention are particularly well suited for use in muscle; whereas vectors based on rh.10 (44-2) capsids of the invention are particularly well suited for use in lung. Uses of such vectors are not so limited and one of skill in the art may utilize these vectors for delivery to other cell types, tissues or organs. Further, vectors based upon other capsids of the invention may be used for delivery to these or other cells, tissues or organs.

Ex. 1007 ¶¶ 141–142.

In Example 9, rh.10 or clone 44.2 was studied for tissue tropism in lung, liver, and muscle tissue. Ex. 1007 ¶¶ 252–262. It was concluded that the "novel AAV capsid clone, 44.2, was found to be a very potent gene transfer vehicle in all 3 tissues with a big lead in the lung tissue particularly." *Id.* ¶ 253. The data from two additional experiments confirmed the "superb tropism" of clone 44.2 in lung-directed gene transfer. *Id.* ¶ 256. The '772 Publication concluded that "[i]nterestingly, performance of clone 44.2 in liver and muscle directed gene transfer was also outstanding, close to that of the best liver transducer, AAV8 and the best muscle transducer AAV1, suggesting that this novel AAV has some intriguing biological significance." *Id.* ¶ 257.

ii. Snowdy (Ex. 1009)

Snowdy discusses nuclear targeting by AAV capsid proteins in the latent phase of the AAV life cycle. *See* Ex. 1009, 31.[6] Snowdy states: "Upon initial infection, the viral genome site-specifically integrates into the host genome, where it remains latent until rescued by one of several helper viruses." *Id.* at 31 (citations omitted). Snowdy notes that this site-specific integration is unique among other latent eukaryotic viruses. *Id.*

Snowdy teaches that it is accepted that AAV capsid proteins target the nucleus, but the mechanism by which they do so is not clear. Ex. 1009, 4, 41. Snowdy looks at a basic region in the AAV capsid including the amino acid sequence ^{307}RPKRLN^{311}, and determines that it plays a role in concentrating AAV capsid proteins in the nucleus during assembly of progeny virions. *Id.* at 5. With respect to a second, similar sequence in the AAV capsid, Snowdy states:

> We mutated several kinase consensus sequences near the AAV ^{166}PARKRLN^{172} region to determine whether the possible phosphorylation sites are important in the infectivity of the virus. We mutated each of three possible kinase phosphorylation targets to either alanine or aspartic acid. We found one site, serine 181, for which transduction efficiency was enhanced by the S181A substitution and for which transduction efficiency was abrogated by converse S181D substitution [that simulates phosphorylation].

Ex. 1009, 5; *see id.* at 128.

In analyzing these results, Snowdy concludes:

> Based on Jans' data for T-ag discussed above, we expected the serine to alanine mutations to result in lowered transduction efficiency if the residue is important for regulating

[6] References are to Petitioner-assigned page numbers.

> the putative NLS [Nuclear Localization Signal]. Likewise, we expected the serine to aspartic acid mutation to suffer, at most, a slight decrease in transduction efficiency versus the total loss of transduction ability that is shown in figure 23. It may be premature in the field's development to assume that a given phosphorylation status will have the same effect on all NLSs, as there is insufficient evidence in the literature to support this assumption. This issue will be better understood with future studies on other nuclear localization signals.

Ex. 1009, 128–129.

Finally, Snowdy notes that "[u]ntil we establish unequivocally that basic region 3 of AAV2 is an NLS, it is premature to assign the cause of the changes in transduction efficiency resulting from mutations to S181 as interfering with the function of an NLS." Ex. 1009, 129. Snowdy, however, assumes that S181 regulates the function of an NLS to speculate on the mechanisms of NLS regulation by S181. Snowdy speculates that "the phosphorylation status of S181 controls the folding of the capsid protein in such a way that results in masking and unmasking of the NLS," or the phosphorylation status "is important in the binding of some cytoplasmic factor that when bound to the capsid protein, causes cytoplasmic retention of the capsid, or interference with the capsid's binding to a nuclear transport factor or the nuclear pore complex." *Id.* at 129.

iii. Fabb (Ex. 1010)

Fabb reports using AAV vectors, which are known to only accommodate less than 5 kb of foreign DNA, with a micro-dystrophin cDNA gene construct that is less than 3.8 kb to restore the dystrophin-associated protein (DAP) complex components and significantly inhibit degenerative dystrophic muscle pathology to treat Duchenne muscular dystrophy. Ex. 1010, 1 (Abstract). Fabb concludes that "[w]e have therefore shown that the current micro-dystrophin gene delivered *in vivo* using an AAV vector is not only capable of restoring sarcolemmal DAP complexes, but can also ameliorate dystrophic pathology at the cellular level." *Id.*

iv. Xie (Ex. 1008)

Xie shows the structure of AAV-2 to 3-Å resolution by x-ray crystallography. Ex. 1008, 1. Xie describes AAV-2 as one of five distinct serotypes with serotypes 1, 2, and 3 sharing about 85% capsid sequence identity and with 2 and 3 likely using the same cellular receptor. *Id.*

From review of the structure of AAV-2, Xie describes the surface topology and assembly. Ex. 1008, 3–4. Xie states:

> The most prominent features are "three-fold-proximal" peaks clustering around each icosahedral three-fold rotation axis. Here, loops from neighboring subunits interact intimately. At the center of each peak is a subloop (VP2 residue 348–379) that is part of the GH loop (between β-strands G and H) of one subunit. It is sandwiched by two other subloops (293–341 and 440–458) from the GH loop of a threefold symmetric subunit (Fig. 4*a*). The subloops of the second subunit are therefore folded away from the β-barrel to extend over a neighboring subunit. Somewhat similarly, a more modest interaction occurs between residues 517 and 531 of the HI loop extending over a fivefold-related neighbor to interact with residues 113–115 and 234–236 of the BC and EF loops, respectively. The loop structures that form these interactions between neighboring subunits are unique to AAV (Fig. 1*c*).

Id. at 4–5.

Xie asserts that with this structure, mapping mutation sites that affect receptor binding that results from different parts of the sequence is possible, and notes that several binding sites "come together in the three-dimensional structure, implicating the side of the threefold-proximal peaks in receptor binding (Figs. 3 and 4)." *Id.* at 5. Xie also concludes:

> The structure also provides a framework for rational site-directed mutagenesis, through which the molecular mechanisms of viral-host interactions can be probed. A rational structure-assisted approach should accelerate the progress in mapping functional sites and modulating function, that has, to date, been attempted through scanning mutagenesis. This mapping, in turn, will provide a foundation for the engineering of recombinant AAV for the more efficient delivery of gene therapies to specifically targeted cells.

Id.

v. Asserted Obviousness Over the '772 Publication and Fabb Alone or in Combination with Snowdy or Xie

Although claim 1 has been disclaimed, claim 8 ultimately depends from claim 1. Therefore, we will discuss the Petitioner's showing with regard to claim 1 as well as claim 8, which remains challenged here.

1. Petitioner's Arguments

Petitioner states that the '772 Publication teaches "[a] recombinant adeno-associated virus (AAV) comprising an AAV capsid and a minigene having AAV inverted terminal repeats and a heterologous gene operably linked to regulatory sequences which direct expression of the heterologous gene in a host cell, wherein the AAV capsid comprises AAV vp1 proteins, AAV vp2 proteins, and AAV vp3 proteins." *See* Pet. 22–25, 47 (citing Ex. 1007 ¶¶ 23, 24, 80, 86, 90, 92, 93, 97, 99, 139, 141; Ex. 1005 ¶¶ 198–200); Ex. 1005 ¶¶ 195–202.

Petitioner also states that the ’772 Publication teaches the limitation, “wherein the AAV vp1 proteins have i) the sequence of amino acids 1 to 738 of SEQ ID NO: 4 (AAVrh46), or ii) an amino acid sequence at least 95% identical to the full length of amino acids 1 to 738 of SEQ ID NO: 4.” Pet. 25–26. According to Petitioner,

> the ’772 Publication discloses the amino acid sequence of the vp1 capsid protein for the preferred embodiment, AAVrh.10, at SEQ ID NO: 81. Alignment of SEQ ID NO:81 (AAVrh.10) from the ’772 Publication with SEQ ID NO: 4 (AAVrh.46) from the ’274 patent using Clustal at default settings shows that the percent identity for these two sequences is 97.29%--which is greater than the “at least 95% identical” threshold in claim 1.

Pet. 25–26 (citing Ex. 1007 ¶¶ 98–103, 264–266, Table 1 (SEQ ID NO: 81, also referred to as “rh.10” and “clone 44.2”), 112–113, Fig. 2; Ex. 1018, 1–2; Ex. 1005 ¶¶ 204–205). Dr. Schaffer testifies that:

> I have aligned SEQ ID NO: 81 (AAVrh.10) from the ’772 Publication with SEQ ID NO: 4 (AAVrh.46) from the ’274 patent using Clustal Omega at default settings. Sequence Alignment of AAVrh.46 (SEQ ID NO: 4) from the ’274 Patent and AAVrh.10 (SEQ ID NO: 81) from the ’772 Publication (EX1018), pp. 1-2. As shown in the alignment, Clustal reports that the percent identity for these two sequences is 97.29% – which is greater than the “at least 95% identical” threshold recited in claim 1. EX1018 (AAVrh.10/AAVrh.46 Alignment), pp. 1-2.

Ex. 1005 ¶ 205.

For the final limitation of claim 1—the amino acid residue corresponding to position 665 in SEQ ID NO: 4 is N when aligned along the full length of amino acids 1 to 738 of SEQ ID NO: 4—Petitioner points to statements in the ’772 Publication that AAVrh.10 has “superb tropism” in lung-directed gene transfer, *see* Ex. 1007 ¶ 256; Ex. 1005 ¶ 207, and AAV8, which has asparagine (N) at position 665, provides "a substantial advantage over the other serotypes in terms of efficiency of gene transfer to liver, *see* Ex. 1007 ¶ 250; Ex. 1005 ¶ 207." Pet. 26. Petitioner also points to statements in the '772 Publication that compare the performance of AAVrh.10 and AAV8, finding AAVrh.10 had outstanding performance in both liver and muscle directed gene transfer, which was close to the best liver transducer, AAV8. Pet. 26–27.

Petitioner reasons:

> The '772 Publication thus teaches the superiority of AAVrh.10 overall and in lung in particular, the superiority of AAV8 in liver, and the importance of efficient gene transfer to the liver for a variety of gene therapy applications. Given these teachings and the express comparison between AAVrh.10 and AAV8 in the '772 Publication, a POSA would have considered substitutions between AAVrh.10 (best overall and best in lung) and AAV8 (best in liver) as a promising strategy for obtaining an artificial variant of AAVrh.10 with even more efficient gene transfer in liver. EX. 1005 ¶ 210.
>
> Moreover, given the already superior performance of AAVrh.10 in lung, a POSA would not have sought to make sweeping substitutions throughout the AAVrh.10 sequence, such as changing multiple amino acids at once, for fear of damaging or destroying the desirable properties of AAVrh.10. Instead, a POSA would have taken a finer, more directed approach, modifying a single amino acid residue at a time. Ex. 1032, 5–6, Table 1; EX. 1005 ¶ 211.
>
> A POSA would have understood from the alignment in Figure 2 of the '772 publication that AAVrh.10 and AAV8 differ at only 48 positions. EX1007, 98-103, Fig. 2, [0071]. EX1019 reproduces the alignment of AAVrh.10 and AAV8 using Clustal O at default settings, similar to the Clustal W program used to create the alignment in Figure 2. EX1019, 1; EX1007, [0071]; EX1005, ¶212.

> One of the differences between AAV8 and AAVrh.10 is at position 665, where AAV8 has an asparagine (N), and AAVrh.10 has a serine (S). EX1019, 1. Thus, based on the teaching in Figure 2 and the experimental data regarding AAVrh.10 and AAV8 disclosed in the Examples, a POSA would have been motivated to make this single amino acid change in AAVrh.10 – namely, substituting an N for the S at position 665. EX1005, ¶213; EX1007, [0074], [0075] ("An artificial AAV serotype may be, without limitation, a chimeric AAV capsid, a recombinant AAV capsid, or a "humanized" AAV capsid.").

Pet. 27–28.

Petitioner offers testimony from one of Patent Owner's declarants, Dr. Paola Leone, in another litigation to confirm this reasoning. *See* Pet. 28–34 (citing Ex. 2012). Petitioner further asserts that analyzing the differences between AAVrh.10 and AAV8 and then making and testing the 48 variants of AAVrh.10, with each containing a single substitution, "was well within the skill of a POSA at the time, and required no more than routine experimentation." Pet. 35 (citing Ex. 1007 ¶¶ 18, 72; Ex. 1005 ¶¶ 227–229).

Petitioner concludes:

> Moreover, a POSA would have had a reasonable expectation of success that a variant of AAVrh.10 with an S to N substitution at position 665 would form a stable AAV capsid capable of packaging a minigene. EX1005, ¶232. According to Patent Owners' expert, Dr. Leone, a POSA would have known with a "reasonable degree of probability" that modified AAVrh.10 capsid sequences having only a small number of mutations – such as a single substitution according to the alignment in Figure 2 – would result in proteins that "may be used to form an AAV capsid." EX1012, ¶¶323, 334, 406.

Pet. 36.

2. Patent Owner's Arguments

Patent Owner counters that Petitioner's reasoning is fatally flawed because each ground in the Petition relies on an argument that a POSITA would take a known capsid protein, AAVrh10, disclosed as one of over fifty different AAV sequences in the '772 publication, as a starting point, modify it by comparing it to one of the other more than fifty sequences disclosed in the '772 publication, AAV8, at just one of the nearly fifty positions where these sequences differ. PO Resp. 1. Patent Owner concludes that because the Petition focuses solely on improving targeting to the liver, the grounds challenging claim 8 fail, "as the Petition is devoid of any motivation that would allow targeting of muscle tissue." *Id.* at 2, 11–12.[7]

In addressing Petitioner's assertion that a POSITA would be motivated to increase AAVrh10's tropism to the liver while preserving its tropism to muscle and lung, Patent Owner states that:

> [I]mproving tropism to the liver, modifying a vector to "preserve" its tropism in multiple tissues, and targeting a therapy to muscle are each different goals, and each would have led a POSITA to different solutions. Critically, none of these solutions would result in the AAV vector claimed in the '274 Patent as the Petition advocates.

PO Resp. 13 (citing Ex. 2027 ¶ 70).

Patent Owner identifies at least four distinct issues with Petitioner's reasoning. First, Patent Owner asserts that the disclosure in paragraph 250 of the '772 Publication does not support Petitioner's assertion that "it was particularly important to identify serotypes with good 'efficiency of gene

[7] Patent Owner also asserts that "all evidence shows that to confer targeting to a specific tissue, a POSITA would change or add a section or fragment of the capsid protein," not make a single amino acid change. PO Resp. 2–3.

transfer to liver that until now has been relatively disappointing." *See* PO Resp. 14 (quoting Pet. 27). Patent Owner asserts that paragraph 250 really states "that gene transfer to liver had been disappointing *until* the discovery of *AAV8*," pointing to the 10 to 100-fold improvement in gene transfer efficiency for the number of hepatocytes stably transduced utilizing AAV8. PO Resp. 14 (emphasis in original) (quoting Ex. 1007 ¶ 250). With this corrected view of the import of paragraph 250, Patent Owner asserts that a POSA, seeking to maximize the efficiency of gene transfer in liver, would choose AAV8 as the starting vector as the logical choice. *Id.* at 15–16 (citing Ex. 2027 ¶ 40–43).

Second, Patent Owner asserts that "Petitioner has cited no evidence that a single amino acid change would be expected to affect tropism, and indeed, the evidence of record—including Petitioner's own evidence [Wu]—shows that a single amino acid change would *not* have been expected to alter tropism." PO Resp. 118 (citing Ex. 2027 ¶¶ 44–48). Patent Owner concludes: "Thus, Petitioner's contention that a POSITA would ignore the '772 Publication's teaching of creating artificial capsids using AAV fragments in favor of site-specific mutagenesis—a technique described nowhere in the '772 Publication for affecting tropism—fails to meet Petitioner's burden." *Id.* at 21.

Third, Patent Owner asserts that in addressing what it deems to be Petitioner's inappropriate new motivation for making changes to AAVrh.10 to target liver tissue, but preserve tropism to muscle and lung tissue, a POSA seeking to improve tropism in one particular tissue would want to *reduce or minimize* tropism to off-target tissues to avoid administering higher doses to deliver an effective amount to the tissue of interest. PO Resp. 23 (citing

Ex. 2027 ¶¶ 31–33, 35, 54; Ex. 2037, 3–4). Patent Owner asserts that specifically for any non-liver tissue target, a POSA would seek to avoid targeting the liver because it is a "default destination" for gene therapies and a "barrier" for targeting non-liver tissues. *Id.* at 24–25 (citing Ex. 2027 ¶¶ 35–36, 55; Ex. 1030, 3). Patent Owner concludes that a POSA "in 2005 would ***either*** have wanted to improve tropism to liver *or* preserve tropism to muscle and lung, not both." *Id.* at 25 (citing Ex. 2027 ¶¶ 55–57).[8]

Finally, with regard to challenged claim 8 requiring that "the dystrophin gene product is a mini-dystrophin or micro-dystrophin," the Patent Owner asserts that "[g]iven the mini-dystrophin or micro-dystrophin are genes that must be delivered to *muscle* tissue, the Petition fails to sufficiently explain why a POSITA, viewing the '772 Publication in view of Fabb, would seek to maximize *liver* function to supply a *muscle*-based dystrophin gene product by modifying AAVrh10 in view of a *liver*-based AAV." PO Resp. 64; *see* PO Resp. 30 ("Dystrophin is a protein predominantly found in muscle tissue, including skeletal muscle, cardiac muscle, and smooth muscle.") (citing Ex. 2027 ¶¶ 60–61; Ex. 1010, 1–6; Ex. 2040, 4–5; Ex. 2041, 1); PO Resp. 31 (stating "it is well understood in gene therapy that *dystrophin* is a desirable protein to deliver to muscles of

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[8] Patent Owner refutes Petitioner's use of Dr. Leone's written description and enablement analysis for claims for a patent related to the '274 patent, stating such opinions address the scope of the disclosure regarding "conserved" and "non-conserved" regions of capsid protein sequences and to identify which amino acid positions could be changed and in what ways to maintain formation of a capsid. PO Resp. 26–27 (citing Ex. 1012 ¶¶ 310–11). Patent Owner states her opinions do not address obviousness or whether a POSA would have made such changes. *Id.*

patients suffering from muscular *dystrophy*, including Duchenne muscular dystrophy”).

Patent Owner posits that a POSITA desiring to target muscle would have identified AAV7 as the vector with the strongest expression in muscle tissue as described in the ’772 publication. PO Resp. 32–33 (citing Ex. 1007 ¶¶ 142, 251; Ex. 2027 ¶ 65). Even if a POSITA would have started with AAVrh10, based on Petitioner’s logic, Patent Owner asserts that the POSITA “would have sought to make changes based on AAV7, as the vector showing the best tropism to muscle, not AAV8, which has weak tropism to muscle.” *Id.* at 34 (citing Ex. 1007 ¶ 244, Table 6; Ex. 2027 ¶¶ 66–68). A comparison of AAVrh10 and AAV1 or AAV7, Patent Owner asserts, would not “lead a POSITA to replace AAVrh10’s serine at position 665 with an asparagine.” *Id.* at 65 (citing Ex. 2027 ¶¶ 68–69).

Patent Owner concludes:

> Petitioner only gets to an asparagine at position 665 by modifying AAVrh10 based on AAV8. But AAV8 was reported in the ’772 Publication to be a poor muscle transducer that a POSITA in 2005 would not have selected for gene delivery in muscle.
>
> Ultimately, the Petition does not address why a POSITA in 2005, starting with the ’772 Publication and looking to target a therapy delivering a dystrophin gene product (thus targeting muscle tissue) would modify an already-promising AAV serotype (AAVrh10) by looking at a poor muscle transducer (AAV8) rather than a higher-performing example (e.g., AAV1 or AAV7).

PO Resp. 65 (citing Ex. 1007 ¶ 244 (for AAV8, showing 4.0 ± 0.7 expression in muscle, compared to 76.0 ± 22.8 in liver).

3. Analysis

We agree with Patent Owner. Petitioner has failed to prove by a preponderance of the evidence that claim 8 would have been obvious over the ’772 Publication and Fabb alone or in combination with Snowdy or Xie because Petitioner has not provided a sufficient motivation to combine the teachings of any of these combinations with a reasonable expectation of success.

We begin our analysis with a review of the requirements of claim 8. Claim 8 claims a recombinant AAV according with all of the limitations of claim 1 (and claim 6), but “wherein the dystrophin gene product is a mini-dystrophin or micro-dystrophin.” We recognize that claim 8 does not require any particular tropism to specific cells or the treatment of any particular disease. *See* Tr. 7:17–8:14, 4:25–15:26. As Petitioner states, it only has to show that a POSA would have been motivated to make the construct of claim 8 regardless of any particular use. Tr. 15:14–17. The question for us to address is whether a POSA would have had reason to make the construct of claim 8 at the time of the invention. Assuming that a POSA would select AAVrh10 as the recombinant AAV to begin with, as Patent Owner points out, the only motivation offered in the Petition for modifying AAVrh10 to arrive at the claimed construct of claim 1 with the amino acid residue at position 665 being asparagine is to improve efficient gene transfer in liver. *See* Pet. 26–27.

The Petition states:

> The ’772 Publication compares the performance of AAVrh.10 to AAV8. “Interestingly, performance of clone 44.2 [AAVrh.10] in liver and muscle directed gene transfer was also outstanding, close to that of the best muscle transducer AAV1,

> suggesting that this novel AAV has some intriguing biological significance." The '772 Publication also teaches that it was particularly important to identify serotypes with good "efficiency of gene transfer to liver that until now has been relatively disappointing . . . ."
>
> The '772 Publication thus teaches the superiority of AAVrh.10 overall and in lung in particular, the superiority of AAV8 in liver, and the importance of efficient gene transfer to the liver for a variety of gene therapy applications. Given these teachings and the express comparison between AAVrh.10 and AAV8 in the '772 Publication, a POSA would have considered substitutions between AAVrh.10 (best overall and best in lung) and AAV8 (best in liver) as a promising strategy for obtaining an artificial variant of AAVrh.10 with even more efficient gene transfer in liver.

Pet. 26–27 (citing Ex. 1007 ¶¶ 250, 252–256; Ex. 1005 ¶ 210).

Dr. Schaffer confirmed that this alleged motivation in the '772 Publication is what he relied upon to show that claim 8 would have been obvious. Dr. Shaffer states: "My opinion [set out in my original declaration] is that a POSA at the relevant time, given the guidance of the '772 Publication and Snowdy, would have been motivated to create a single amino acid mutation in the AAVrh.10 capsid to try to confer some of the favorable liver tropism of AAV8 onto AAVrh.10." Ex. 1072 ¶ 9.

The '772 Publication, however, does not indicate as Petitioner asserts that a new variant is necessary to improve the efficiency of gene transfer to liver; the '772 Publication indicates that AAV8 fulfills that need. *See* Ex. 1007 ¶ 250. In the paragraph that Petitioner relies upon for the alleged need to improve the efficiency of gene transfer to the liver, the '772 Publication states in full:

> The tropism of each new vector is favorable for in vivo applications. AAV2/7 vectors appear to transduce skeletal

> muscle as efficiently as AAV2/1, which is the serotype that confers the highest level of transduction in skeletal muscle of the primate AAVs tested to date. *Importantly, **AAV2/8 provides a substantial advantage over the other serotypes** in terms of efficiency of gene transfer to liver that until now has been relatively disappointing in terms of the numbers of hapatocytes stably transduced*. AAV2/8 consistently achieved a 10 to 100-fold improvement in gene transfer efficiency as compared to the other vectors. The basis for the improved efficiency of AAV2/8 is unclear, although it presumably is due to uptake via a different receptor that is more active on the basolateral surface of hepatocytes. ***This improved efficiency will be quite useful in the development of liver-directed gene transfer where the number of transduced cells is critical, such as in urea cycle disorders and familial hypercholesterolemia***.

Id. (citations omitted) (emphasis added).

According to the import of the complete recitation of paragraph 250 of the ’772 Publication set forth above that Petitioner relies upon for the motivation to improve efficient gene transfer in liver, AAV8 already *fulfills* this need. *See* Tr. 28:15–18. We do not read the ’772 Publication as calling for a search for a more efficient variant to improve efficient gene transfer in liver. We credit Dr. Samulski’s testimony that supports this view of the teachings of the ’772 Publication. Dr. Samulski testifies:

> Additionally, when the inventors compared AAV8's expression in the liver against new serotypes AAVrh2, AAVrh10, AAVrh13, and AAVrh21, they found that AAV8 remained superior:
>
> TABLE 8
>
> | | Target Tissue | | |
> |---|---|---|---|
> | Vector | Lung | Liver | Muscle |
> | AAV2/1 | ND | ND | 45 ± 11 |
> | AAV2/5 | 0.6 ± 0.2 | ND | ND |
> | AAV2/8 | ND | 84 ± 30 | ND |
> | AAV2/rh.2 (43.1) | 14 ± 7 | 25 ± 7.4 | 35 ± 14 |
> | AAV2/rh.10 (44.2) | 23 ± 6 | 53 ± 19 | 46 ± 11 |
> | AAV2/rh.13 (42.2) | 3.5 ± 2 | 2 ± 0.8 | 3.5 ± 1.7 |
> | AAV2/rh.21 (42.10) | 3.1 ± 2 | 2 ± 1.4 | 4.3 ± 2 |
>
> EX1007, ¶ 253. Notably, AAV8 was shown to have over fifty percent (50%) greater expression in liver than AAVrh10. As I discuss more below, a skilled artisan in 2005 would not have expected a single amino acid substitution to significantly alter tropism in order to target to a different tissue. Thus, in my opinion, in 2005, a skilled artisan with the goal of targeting the liver would have selected the best known vector for delivery to liver at that time—AAV8.

Ex. 2027 ¶ 42.

We also agree with Dr. Samulski's opinion that a POSA would have selected the best vector for liver, AAV8, instead of modifying another vector to increase tropism to liver. Dr. Samulski testifies:

> In my opinion, a skilled artisan in 2005 would not have looked solely to modify a vector to try to improve tropism to a particular tissue. As discussed throughout this declaration, changing tropism is difficult to predict, and a skilled artisan would not have had a reasonable expectation that changes could be made to improve or alter tropism. A skilled artisan would also have been concerned that changes, even small ones, could negatively impact other properties of the capsid protein. Therefore, rather than seek to modify a vector, a skilled artisan

> in 2005 instead would have merely selected the best available vector for the tissue she wanted to target. If that tissue was liver, a skilled artisan in 2005 would have chosen AAV8.

Ex. 2072 ¶ 43.

Because we find that the motivation to improve efficient gene transfer in liver is not supported by the record before us, we also find that Petitioner has failed to show a motivation to compare AAVrh.10 to AAV8 to attempt to improve the ability of AAVrh.10 for efficient gene transfer in liver. We find the Petitioner is using impermissible hindsight to arrive at the invention in claim 8. *See Mintz v. Dietz & Watson, Inc.*, 679 F.3d 1372, 1377 (Fed. Cir. 2012) (hindsight to use the invention to define the problem the invention solves). This hindsight is particularly acute in the present case, where a number of uncertain selections would have been necessary including particular starting AAV sequences and particular point mutations, with no direct evidence that those alterations would have necessarily resulted in improved liver tropism. Due to hindsight alone, we determine that Petitioner has failed to prove by a preponderance of the evidence that claim 8 is unpatentable.

We will further address Petitioner's reasoning concerning how claim 8 would have been obvious over the '772 Publication and Fabb alone or in combination with nowdy or Xie that it raised for the first time in its Reply. Petitioner asserts that a POSA seeking to use the variant of AAVrh.10 with the claimed substitution at position 665 "as a delivery vehicle for a dystrophin gene would have been motivated to combine the '772 Publication and Snowdy with Fabb, which teaches the construction of a smaller version of dystrophin gene known as a 'micro-dystrophin' for use with an AAV vector" with a reasonable expectation of success. Pet. 58–59 (citing Ex. 1010, 2–3; Ex. 1005 ¶¶ 305–307).

In our Institution Decision, we noted that:

> Patent Owner does not question Fabb's teachings as set forth by Petitioner at this point, but responds that this ground fails because claim 8 requires tropism to *muscle* tissue, not to lung or liver, a fact Petitioner does not address. Prelim. Resp. 57. Patent Owner states "the Petition is silent as to why an AAV that was modified to have alleged improved liver tropism would have been used to carry a mini or micro-dystrophin and target muscle." *Id*. at 59.
>
> We note that the '772 Publication states that "performance of clone 44.2 in liver and muscle directed gene transfer was also outstanding" indicating that AAVrh.10 (at least before modification) performed well with muscle tissue, but invite the parties to address this issue further at trial. *See* Ex. 1007 ¶ 257; *see id*. ¶ 253.

Dec. 28.

In the Reply, Petitioner provides further motivation for a POSA to make the construct of claim 8 that has a mini-dystrophin or micro-dystrophin gene product that would be produced generally in muscle byenhancing the motivation originally provided in the Petition to include targeting muscle. *See* Reply 15–20.

Originally, Dr. Schaffer testified that a POSA would modify a single amino acid at a time in AAVrh10 because

> given the already *superior performance of AAVrh.10 in lung*, a POSA would not have sought to make sweeping substitutions throughout the AAVrh.10 sequence, such as changing multiple amino acids at once, *for fear of damaging or destroying the desirable properties of AAVrh.10*. Instead, a POSA would have taken a finer, more directed approach, modifying a single amino acid residue at a time. See, e.g., EX1032 (Wu), pp. 5-6, Table 1.

Exhibit 1005 ¶ 211 (emphasis added).

Petitioner asserts that we should read this testimony of Dr. Schaffer as indicating a POSA would have modified AAVrh.10 to improve liver tropism while also preserving AAVrh.10's superior tropism in lung *and* muscle. From Dr. Shaffer's statement above that a POSA would modify AAVrh.10 one amino acid at a time "for fear of damaging or destroying the desirable properties of AAVrh.10," Petitioner elaborates further on the motivation to alter AAVrh.10. Petitioner states:

> A POSA, given the teachings of the '772 Publication about the interesting properties of AAVrh.10 – specifically, its superior transduction across three tissues – would also have been guided by the express teachings of the '772 Publication that there was a particular need in the art for AAV vectors with improved liver transduction. EX1007, [0250]; EX1005, ¶¶137-38; EX1072, ¶¶351-61, 394-409.
>
> Importantly, however, improving liver transduction while destroying the superior lung and muscle tropism would defeat the entire purpose of modifying AAVrh.10 in the first place, and would ignore the teachings of the '772 Publication about the importance of the superiority of AAVrh.10 across multiple tissues. EX1007, [0257]; EX1072, ¶358.

Reply 16–17.

Dr. Shaffer's testimony in his original declaration submitted in support of the Petition, however, referred only to the preservation of the superior performance of AAVrh.10 *in lung* with no mention of muscle at all in his original testimony. We find that Petitioner has changed its proffered motivation for a POSA to make the construct of claim 8. It is untimely for

Petitioner to offer a new motivation in its Reply[9]; Patent Owner should not have to respond to a moving target to defend the patentability of its claims.

Because we find that Petitioner has not shown sufficiently why a POSA would modify AAVrh.10 to arrive at the construct of claim 8 based on the combination of ’772 Publication and Fabb, which applies to all three grounds challenging claim 8, we need not reach issues raised by the parties concerning Snowdy and Xie that do not address this lack of motivation to combine the teachings of the references to arrive at claim 8. *See* Tr. 43–44.

IV. CONCLUSION

For the foregoing reasons, we conclude Petitioner has not shown by a preponderance of the evidence that claim 8 of the ’274 patent is unpatentable.

V. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that claim 8 of U.S. Patent No. 11,680,274 B2 has not been shown to be unpatentable under 35 U.S.C. § 103(a); and

FURTHER ORDERED that because this is a Final Written Decision, parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

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[9] Petitioner may not submit new arguments that it could have presented earlier to make out a prima facie case of unpatentability. *See* Consolidated Trial Practice Guide (*available at* https://www.uspto.gov/TrialPracticeGuideConsolidated) 73.

In summary:

Claims	35 U.S.C. §	Reference(s)/Basis	Claims Shown Unpatentable	Claims Not Shown Unpatentable
8	103(a)	’772 Publication, Fabb		8
8	103(a)	’722 Publication, Xie, Fabb		8
8	103(a)	’722 Publication, Snowdy, Fabb		8
Overall Outcome				8

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